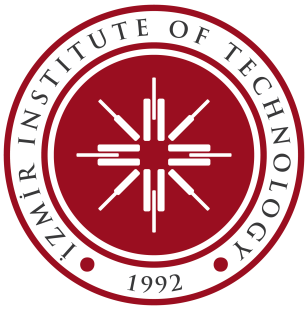


İzmir Institute of Technology

Faculty of Engineering

Department of Electrical and Electronics Engineering

EE 315 Electronics Laboratory Prelab Sheet

Experiment 2

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Preliminary Work

Problem 1. Full-Wave Bridge Rectifier

a) Positive half cycle: D1 and D3
Negative half cycle: D2 and D4

b) 100 Hz because the Full wave rectifier folds the negative voltage to positive so the VL frequency is twice the input.

c) Peak voltage is 9 volts (Vs=9) so according to $2V_{sp}/\pi$ formula the average voltage is 5.730v but because of the diodes which are in series the total voltage drops by 1.2 volts so the average voltage drops to 4.530 volts. The the DC load current can be calculated with V_{LDC}/R_L which is equal to 3.775mA. In the end the total power will be calculated as $2 \cdot V_D \cdot I_{LDC} = 2 \cdot 0.6\text{ V} \cdot 3.775\text{mA} = 4.53\text{mW}$.

Problem 2. Full-wave Rectifier with a Center-tapped Transformer

1. Average DC Load Voltage: $V_{LDC} = 18/\pi - 0.6 = 5.130\text{V}$
2. Average DC Load Current: $I_{LDC} = 5.130/1200 = 4.275\text{ mA}$
3. Total Power: $1 \cdot V_D \cdot I_{LDC} = 0.6 \cdot 4.275 = 2.57\text{ mW}$

Problem 3. Full-wave Bridge Rectifier with Triangular Approximation

a) Load Regulation ($R_L = 330$ ohms)

1. **No-load condition (V_{NL}):** When R is infinity no current is drawn so the capacitor does not charge and voltage remains at peak.
   $V_{NL} = V_{LP} = 7.8\text{V}$
2. **Full Load Condition:** $f_r R_L C = 100 \cdot 330 \cdot 22 \cdot 10^{-6} = 0.726$
   $V_{FL} = 7.8/(1 + (1/(2 \cdot 0.726))) = 4.619\text{V}$
3. **Calculate Load Regulation:**
4. Load Reg $= \left(\frac{V_{NL} - V_{FL}}{V_{NL}}\right) 100\% = \left(\frac{7.8 - 4.619}{7.8}\right) \cdot 100\% = 40.78\%$

b) Calculate Ripple Factor

- **For Load Resistance = 330 ohms:**
  Ripple Factor = $1/(3.4641 \cdot 100 \cdot 330 \cdot 0.000022)$
  Ripple Factor = 0.3976
  Result = 39.76%
- **For Load Resistance = 1.2 kohms (1200 ohms):**
  Ripple Factor = $1/(3.4641 \cdot 100 \cdot 1200 \cdot 0.000022)$
  Ripple Factor = 0.1093
  Result = 10.93%
- **For Load Resistance = 10 kohms (10000 ohms):**
  Ripple Factor = $1/(3.4641 \cdot 100 \cdot 10000 \cdot 0.000022)$
  Ripple Factor = 0.0131
  Result = 1.31%

c) Effect of Load Resistance on Ripple Voltage:
As the load resistance increases, the load draws less current from the circuit. A lower current draw means the filter capacitor discharges much slower between the active charging cycles. Because the capacitor loses less

stored voltage before the next pulse arrives, the peak-to-peak voltage drop (the ripple) is significantly reduced, resulting in a cleaner, smoother DC output.

Problem 3. Full-Wave Bridge Rectifier without Triangular Approximation

Because the time constant $\tau = R_L C_1 = 330\,\Omega \times 22\,\mu\text{F} = 7.26$ ms is shorter than the 10 ms ripple period, the capacitor discharges deeply. The standard triangular ripple approximation ($V_{rplPP} = \frac{V_{LDC}}{fR_LC}$) fails under these large-ripple conditions. We must use the exact exponential discharge method.

a) Load Regulation
The peak load voltage is $V_{LP} = 9\text{ V} - 2(0.6\text{ V}) = 7.8$ V. At no-load ($R_L \to \infty$), the capacitor does not discharge, so $V_{NL} = V_{LP} = \mathbf{7.8}$ V.

At full load ($R_L = 330\,\Omega$), the capacitor voltage decays exponentially from the peak at $t = 0$:

$$v_C(t) = V_{LP} e^{-t/\tau} = 7.8 e^{-t/7.26\text{ ms}}$$

The rectifier diodes turn back on when this decaying voltage intersects the rising edge of the next rectified half-sine wave, given by $v_{\text{rect}}(t) = -9\cos(100\pi t) - 1.2$. Setting them equal:

$$7.8 e^{-t/7.26\text{ ms}} = -9\cos(100\pi t) - 1.2$$

Solving this transcendental equation numerically yields an intersection time of $t \approx 6.62$ ms. Substituting this back into the discharge equation gives the minimum trough voltage:

$$V_{\text{min}} = 7.8 e^{-6.62/7.26} \approx 3.13\text{ V}$$

The average DC voltage is approximately the midpoint of the peak and trough:

$$V_{FL} = V_{LDC} \approx \frac{V_{LP} + V_{\text{min}}}{2} = \frac{7.8 + 3.13}{2} = 5.465\text{ V}$$

$$\%\text{ Load Regulation} = \frac{V_{NL} - V_{FL}}{V_{NL}} \times 100\% = \frac{7.8 - 5.465}{7.8} \times 100\% = \mathbf{29.93}\%$$

b) Ripple Factor
The peak-to-peak ripple voltage is $V_{rplPP} = 7.8\text{ V} - 3.13\text{ V} = 4.67$ V. Using the definition from the manual:

$$\text{Ripple Factor} = \frac{V_{rplPP}}{2\sqrt{3}V_{LDC}} = \frac{4.67}{2\sqrt{3}(5.465)} = \frac{4.67}{18.93} = \mathbf{24.67}\%$$

(Note: As R_L increases to 1.2 kΩ and 10 kΩ, the time constant τ becomes very large, and the standard linear approximation becomes accurate again, yielding $\approx 10.9\%$ and $\approx 1.3\%$ respectively.)

Problem 4. Capacitive Filtering

a) Load Regulation with Added Capacitor

Adding a second $22\,\mu$F capacitor in parallel makes $C_{tot} = 44\,\mu$F. The no-load voltage remains $V_{NL} = 7.8$ V. The new time constant is $\tau = 330\,\Omega \times 44\,\mu\text{F} = 14.52$ ms. Setting the new exponential discharge equal to the rectified wave:

$$7.8e^{-t/14.52\text{ ms}} = -9\cos(100\pi t) - 1.2$$

Solving numerically gives $t \approx 7.28$ ms. The new trough voltage is:

$$V_{\min} = 7.8e^{-7.28/14.52} \approx 4.72\text{ V}$$

$$V_{FL} = V_{LDC} = \frac{7.8 + 4.72}{2} = 6.26\text{ V}$$

$$\%\text{ Load Regulation} = \frac{7.8 - 6.26}{7.8} \times 100\% = \mathbf{19.74}\%$$

b) Effect of Increasing Filter Capacitance

Increasing the filter capacitance directly increases the *RC* time constant. This slows the voltage decay rate during the load discharge phase, preventing the voltage from dipping as low before the next cycle. The result is a significantly reduced ripple voltage (4.67 V down to 3.08 V) and an improved load regulation percentage.

5. LTspice Simulation

All plots and variables are derived from the example circuit provided in the related subsection, which are configured using the component values specified in the laboratory manual.

Note: Related questions are answered at the end of each subsection.

5.1 Used SPICE Models

```
.model 1N750 D(Is=.88f Rs=.25 Cjo=175p M=.55 nbv=1.7 bv=4.7 Vj=.75 Isr=1.86n Nr=2
Ibv=20.245m Ibvl=1.96m Nbvl=15 Tbv1=-21.3u Vpk=4.7 mfg=OnSemi type=Zener)

.model 1N4007 D(Is=90p Rs=40m Cjo=30p N=1.4 TT=5u Iave=1 Vpk=1000 mfg=Diodes_Inc.
type=silicon)
```

5.2 Full-Wave Bridge Rectifier

The purpose of a full-wave bridge rectifier circuit is to convert an AC voltage to a DC voltage. As the bridge rectifier rectifies both half-cycles of the input signal into a single polarity, the frequency of the output voltage is doubled relative to the input frequency. Since two diodes conduct in series during each half-cycle, the circuit has a voltage drop equal to the sum of the forward voltage drops of the two conducting diodes. To prevent severe voltage sag, a filter capacitance is needed. An example circuit is given in Figure 1.

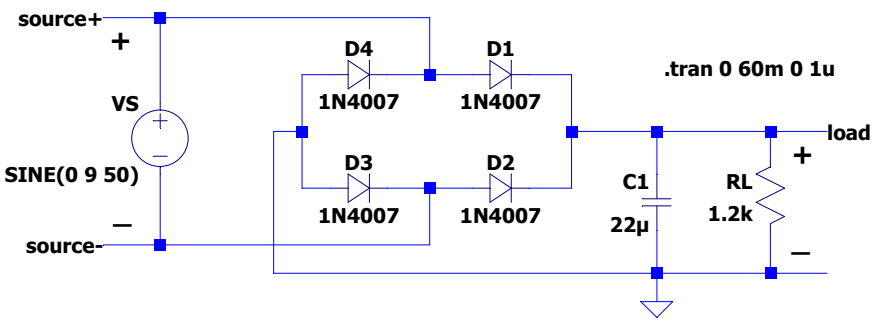


Figure 1: Schematic of full-wave bridge rectifier circuit

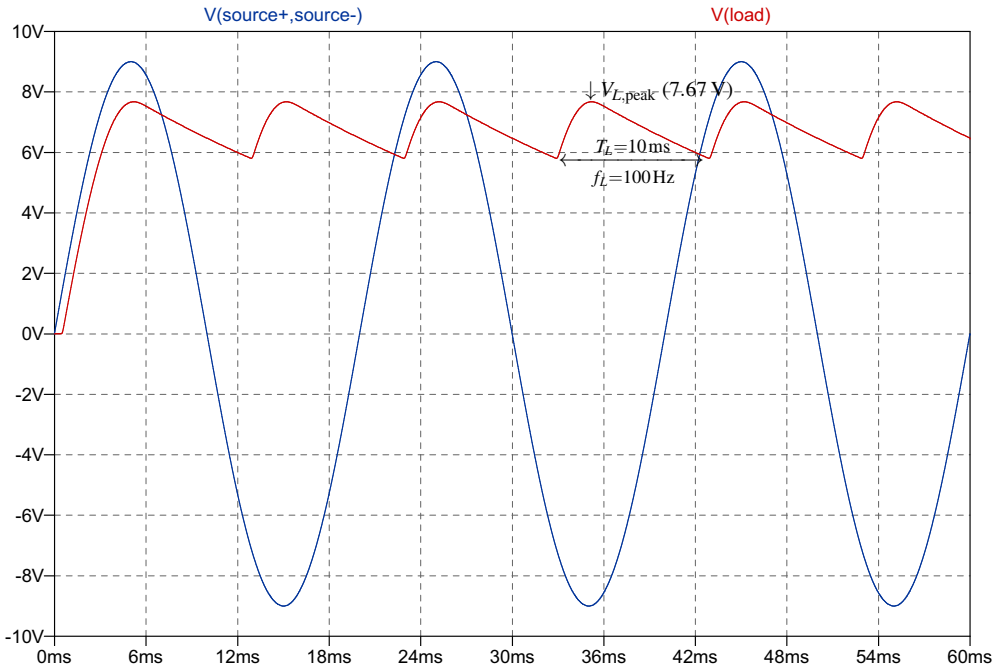


Figure 2: Input and output voltage waveforms of the full-wave bridge rectifier circuit

5.2.1 Questions

a.) *Determine the peak value and frequency of v_L, and compare the results with your calculations in Question-1*

$V_{L,\text{peak}}$	f_L	**Number of conducting diodes**
7.67 V	100 Hz	2

– Output Ripple Frequency (f_L): It is the same for both theoretical calculations and simulation results since both rely on the same frequency-doubling formula ($f_L = 2 \times f_s$).

– Peak Load Voltage ($V_{L,\text{peak}}$): The diode forward voltage drop is greater in the simulation because the model accounts for dynamic and non-linear parameters, whereas the theoretical calculation assumes a constant forward voltage drop. This results in a slightly lower peak load voltage in the simulation compared to hand calculations.

b.) *Include the plot showing v_S and v_L in your prelab report.*

– The input and load voltage of the example circuit is given in Figure 2.

5.3 Zener Diode Regulator

In full-wave bridge rectifier circuits, a key limitation is that the peak output voltage ($V_{L,\text{peak}}$) and the peak-to-peak ripple voltage ($V_{rpl,PP}$) are affected by load conditions. As the load current increases, the voltage drops across the bridge diodes and series components increase, causing the output voltage to sag. To address this limitation and clamp the peak output voltage to a desired, more stable level, a Zener diode regulator can be implemented. However, since the excess voltage is entirely dropped across the series resistor (R_S), this circuit suffers from poor thermal and power efficiency, especially in high-current applications. Despite this inefficiency, it remains a low-cost practical solution for low-current voltage references and basic low-power circuits. An example circuit is given in Figure 3.

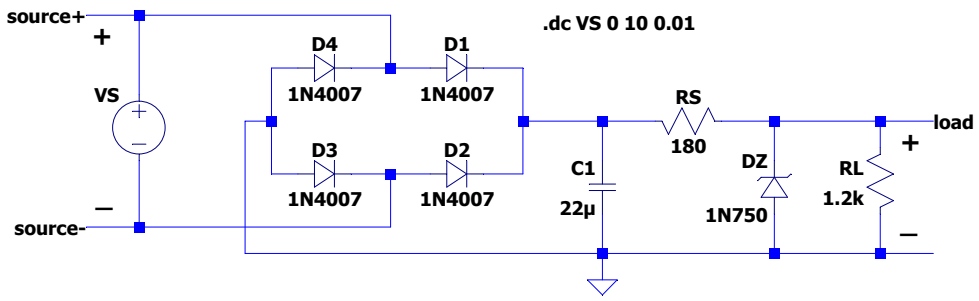


Figure 3: Schematic of Zener diode regulator circuit

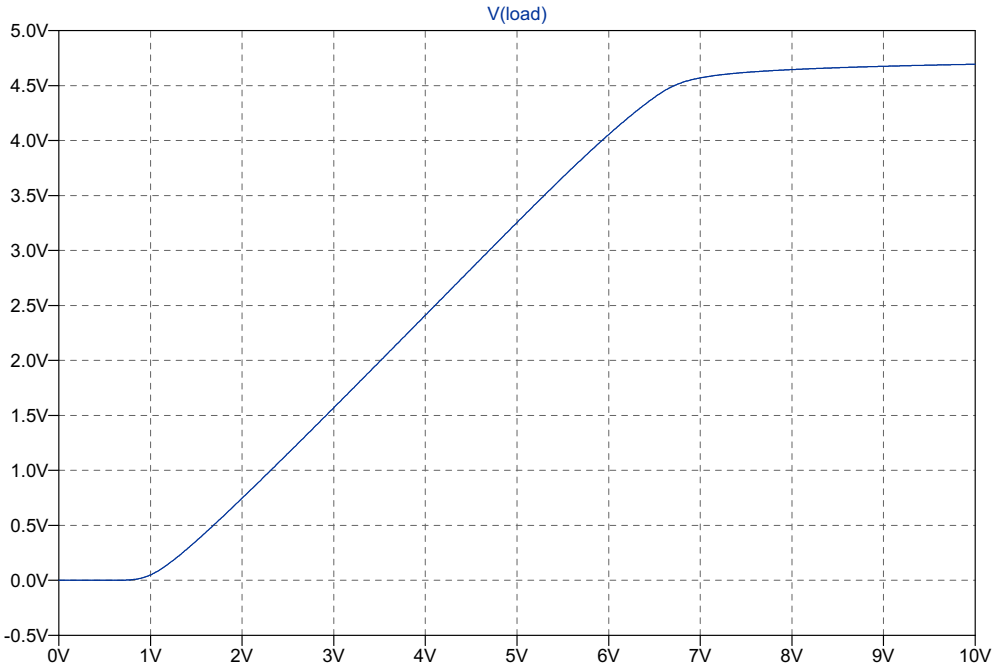


Figure 4: Regulated output voltage (v_L) as a function of the swept input voltage (v_S)

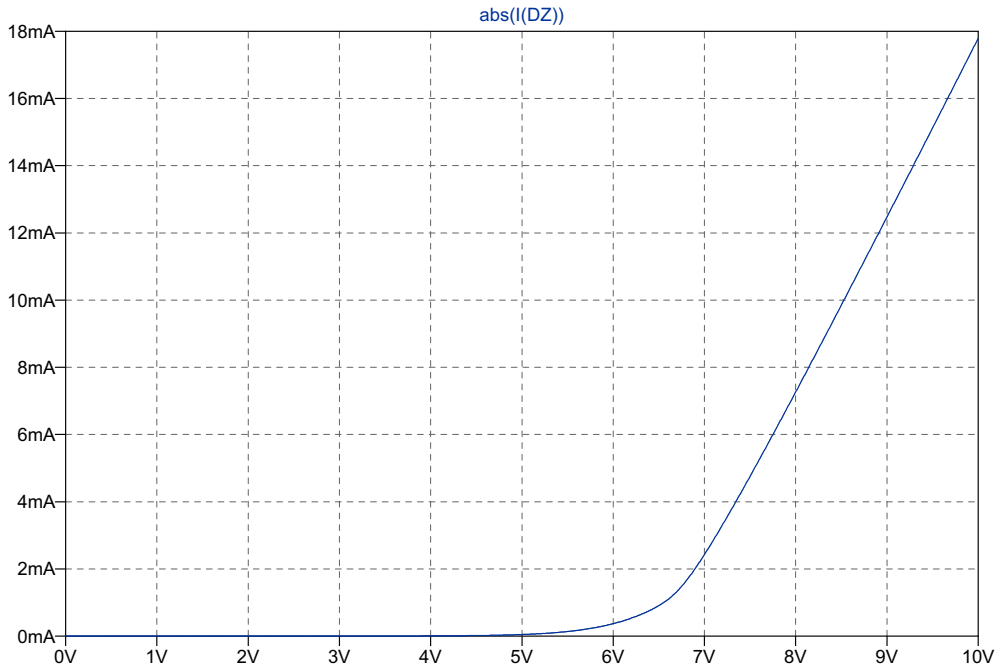


Figure 5: Zener breakdown-current magnitude (i_Z) versus the swept input voltage (v_S)

5.3.1 Questions

a.) *Determine the DC output voltage and the Zener diode current.*

Breakdown Voltage (V_Z)	$V_{o,DC}$	I_Z
4.7 V	2.83 V	3.1 mA

b.) *Determine the approximate input voltage at which regulation begins and the regulated output voltage.*

– The regulation starts at breakdown voltage of the Zener diode ($V_Z = 4.7\,\text{V}$) as the diode limits the output voltage from exceeding it.

c.) *Include the plots of v_L and the Zener breakdown-current magnitude i_Z versus v_C in your prelab report.*

– The mentioned plots of the example circuit are given in Figure 4 and Figure 5.